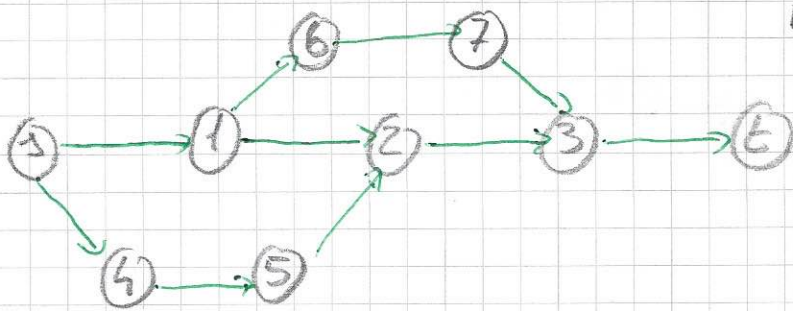
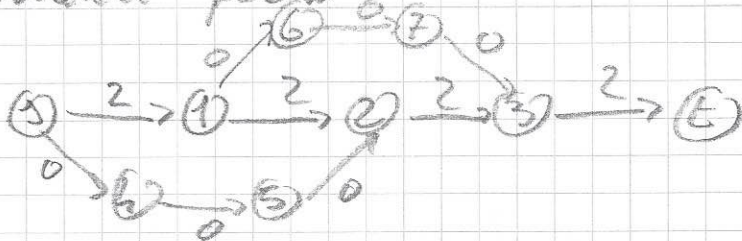


1)

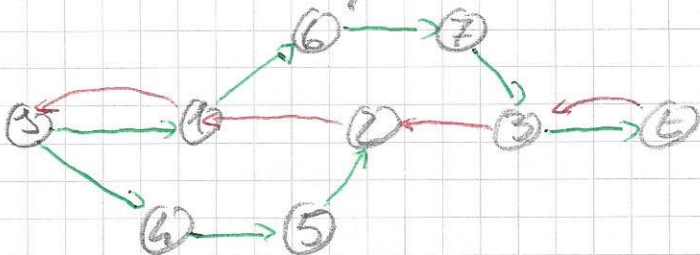
Residual graph $x=0$ Path $s - 1 - 2 - 3 - t$

$$\theta = \min \{ 3-0, 2-0, 2-0, 8-0 \} = 2$$

Updated flow

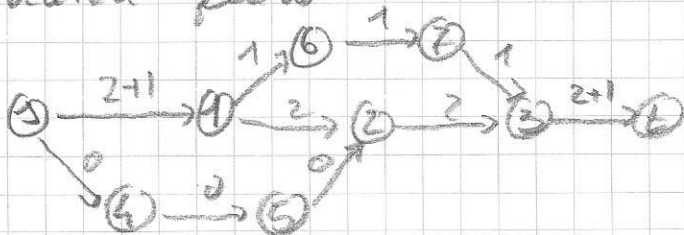


New residual graph

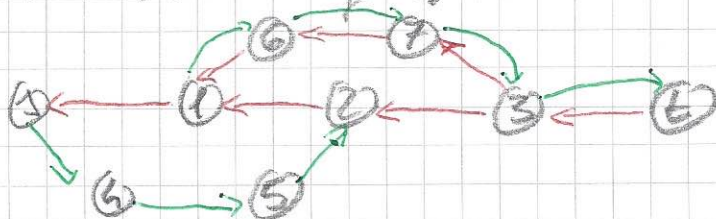
Path $s - 1 - 6 - 7 - 3 - t$

$$\theta = \min \{ 3-2, 6-0, 5-0, 11-0, 8-2 \} = 1$$

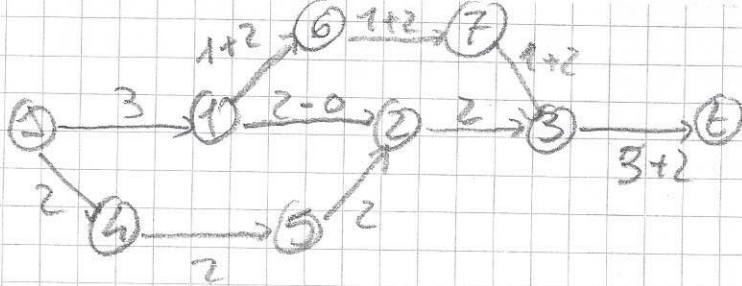
Updated flows



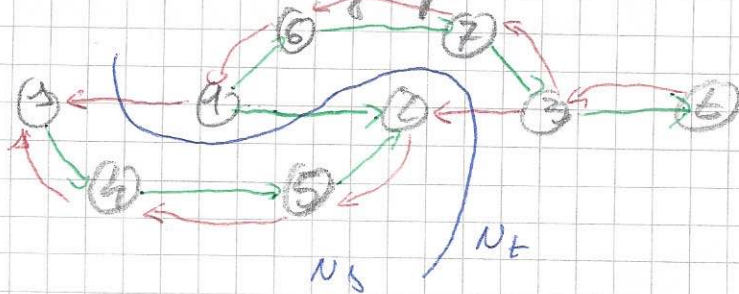
New residual graph

path $s - 4 - 5 - 2 - 1 - 6 - 7 - 3 - t$

$$\theta = \min \{ 3-0, 10-0, 5-0, 2, 6-1, 5-1, 11-1, 8-3 \} = 2$$



New residual graph



Cut $N_S = \{1, 2, 4, 5\}$ $N_T = \{3, 6, 7, 8\}$

Arc $(1, 2)$ goes from N_T to N_S thus it has flow $x_{12} = 0$. Increasing the capacity u_{12} would not change the solution.

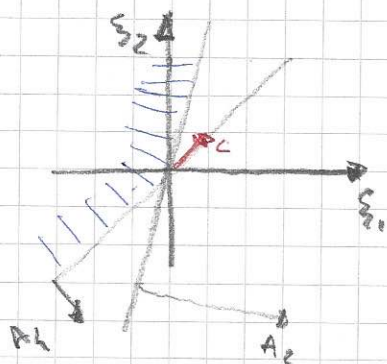
Arc $(2, 3)$ goes from N_S to N_T thus it has flow $x_{23} = u_{23}$. Increasing the capacity u_{23} would increase the cut capacity and potentially also the maximum flow.

2)

$$\bar{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad I(\bar{x}) = \{2, 4\}$$

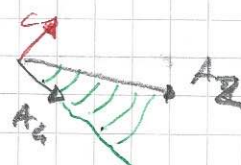
growing feasible direction

$$\begin{cases} \xi_1 + \xi_2 > 0 \\ \xi_1 - \xi_2 \leq 0 \\ \xi_1 - \xi_2 \leq 0 \end{cases}$$

 $\exists$ a solution $\bar{\xi}$

restricted dual

$$\begin{cases} \eta_1 + \eta_2 = 1 \\ -\eta_1 - \eta_2 = 1 \\ \eta_1, \eta_2 \geq 0 \end{cases}$$



no solution

the given $\bar{\xi} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ satisfies the condition $A_2 \bar{\xi} = 0$
 $A_4 \bar{\xi} = -3 < 0$

Moving step computation

$$A_1 \bar{\xi} = \begin{bmatrix} -4 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = 28 > 0$$

$$A_3 \bar{\xi} = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = 8 > 0$$

$$\min \left\{ \frac{b_1 - A_1 \bar{x}}{A_1 \bar{\xi}}, \frac{b_2 - A_2 \bar{x}}{A_2 \bar{\xi}} \right\} = \left\{ \frac{14 - 0}{28}, \frac{5 - 2}{8} \right\} = \frac{3}{8}$$

$$\bar{x} \leftarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{3}{8} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 13/8 \\ 5/2 \end{bmatrix}$$

$$\begin{aligned}
 3) \quad & \max z = x_1 + x_2 \\
 & -2x_1 + 4x_2 + x_3 = 7 \\
 & 6x_1 + 8x_2 + x_4 = 27 \\
 & x_i \geq 0 \quad i = 1 \dots 4
 \end{aligned}$$

$$B = \{2, 4\} \quad A_B = \begin{bmatrix} 4 & 0 \\ 8 & 1 \end{bmatrix} \quad A_B^{-1} = \begin{bmatrix} 1/4 & 0 \\ -2 & 1 \end{bmatrix} \quad A_N = \begin{bmatrix} -2 & 1 \\ 6 & 0 \end{bmatrix}$$

$$A_B^{-1} b_B = \begin{bmatrix} 1/4 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 27 \end{bmatrix} = \begin{bmatrix} 7/4 \\ 13 \end{bmatrix} \rightarrow \begin{array}{ll} \text{fractional} & : \text{cut 1} \\ \text{integer} & : \text{no cut} \end{array}$$

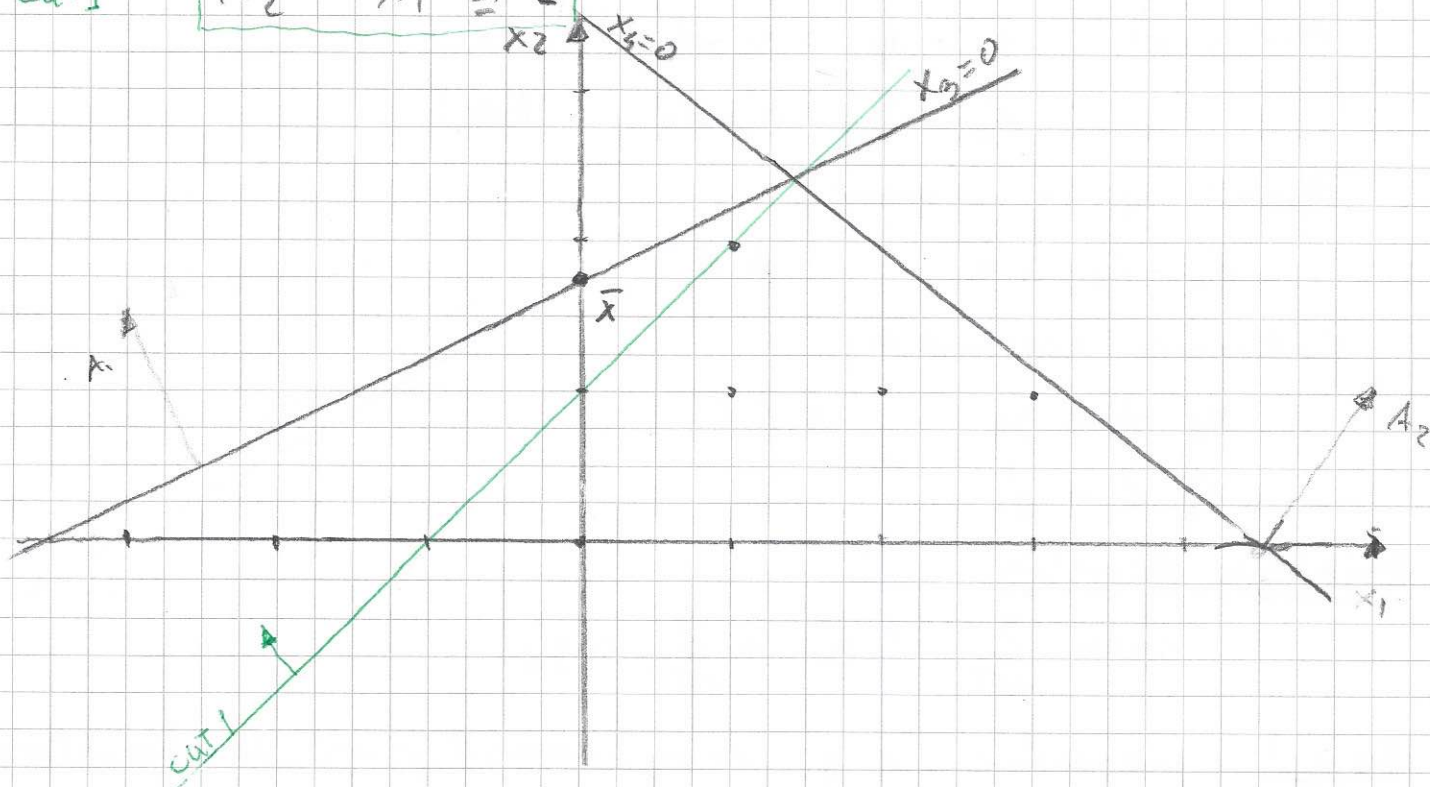
$$A_B^{-1} A_N = \begin{bmatrix} 1/4 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 6 & 0 \end{bmatrix} = \begin{bmatrix} -1/2 & 1/4 \\ 10 & -2 \end{bmatrix} \quad \begin{array}{ll} & x_2 \\ & x_4 \\ x_1 & x_3 \end{array}$$

cut 1

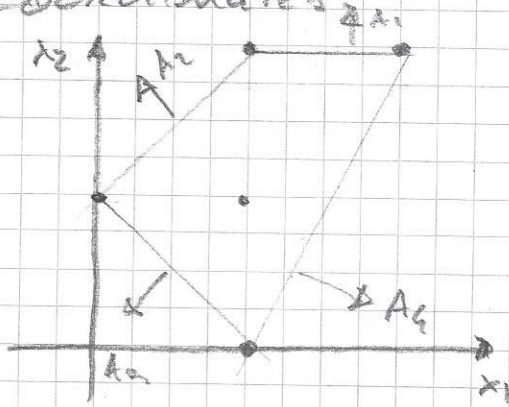
$$1x_2 + \left[-1/2\right]x_1 + \left[1/4\right]x_3 \leq \left[7/4\right]$$

cut 1

$$x_2 - x_1 \leq 1$$



4) The convex hull of F is the smallest convex polyhedron containing all the points of F . If F contains only integer coordinate points of $\mathbb{R}^n$ then the most important property of the convex hull is that its vertices have integer coordinates.



$$x_2 \leq 2$$

$$-x_1 + x_2 \leq 1$$

$$-x_1 - x_2 \leq -1$$

$$2x_1 - x_2 \leq 2$$

5) sets $T = \{1 \dots m\}$ days

Parameters

d_t number of bottles of water used in day t
 d'_t " " wine " " t
 w initial number of bottles of water
 w' " " wine
 c cost of the delivery for water
 c' " " for wine
 u capacity of the basement (number of bottles)

Variables

$x_t \geq 0$ n. of bottles of water bought in day t
 $x'_t \geq 0$ " " wine " " t
 $z_t \geq 0$ n. of bottles of water in the basement
 $z'_t \geq 0$ " wine " "
 $y_t = \begin{cases} 1 & \text{if } x_t > 0 \\ 0 & \text{otherwise} \end{cases}$ $z_0 = w$
 $y'_t = \begin{cases} 1 & \text{if } x'_t > 0 \\ 0 & \text{otherwise} \end{cases}$ $z'_0 = w'$

Constraints

Multiperiod

$$z_t = z_{t-1} + x_t - d_t \quad t \in T$$

$$z'_t = z'_{t-1} + x'_t - d'_t \quad t \in T$$

capacity of the basement

$$z_t + z'_t \leq u \quad t \in T$$

coherence constraint

$$x_t \leq u y_t \quad t \in T$$

$$x'_t \leq u y'_t$$

Objective function

$$\min \sum_{t \in T} c y_t + \sum_{t \in T} c' y'_t$$

Variant

parameters: q, q' min purchased quantity, for free delivery

variables $d_t = \begin{cases} 1 & \text{if the min quantity of water is reached} \\ 0 & \text{otherwise} \end{cases}$
 $d'_t = \begin{cases} 1 & \text{" " wine " " } \\ 0 & \text{otherwise} \end{cases}$

new coherence constraints $d_t q \leq x_t \leq u y_t, d'_t q \leq x'_t \leq u y'_t$

objective function $\min \sum_{t \in T} c(y_t - d_t) + \sum_{t \in T} c'(y'_t - d'_t)$

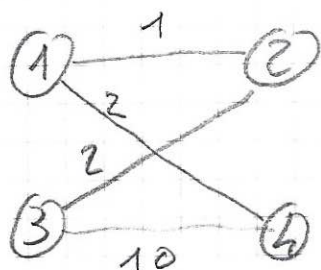
6)

$$E = A$$

$$F = \{ A' \subset A : \text{no two arcs in } A' \text{ are incident in the same node} \}$$

$$\text{Best}(S) = \arg\min \{ c_{ij} \mid (i,j) \in S \}$$

$$\text{Ind}(X, (i,j)) = i \text{ and } j \text{ have no incident arcs}$$



The Greedy selects (1,2) and (3,4)
cost 11

The optimal solution (1,4) and (3,2)
cost 4